

Project Idea: Air Mouse

- Discussed project idea for the AirMouse device.
- The goal is to control a computer cursor using a gyroscope.

Design Decision:

- Decided to use a motion sensor to detect movement.
- Arduino will process the sensor data.

Reason:

- Arduino is easy to program and supports sensor libraries.

Component Selection:

- Researched motion sensors
- Selected MPU6050 accelerometer/gyroscope module.

Design Decision:

- Use I2C communication with Arduino.

Reason

- I2C only requires two communication wires (SDA and SCL).

Circuit Design:

- Built the circuit on a breadboard.
- Connected MPU6050 to Arduino.

Connections:

- VCC - 5V
- GND - GND
- SDA - A4
- SCL - A5

Reason:

- Breadboard allows easy testing and modification of the circuit.

Sensor Communication Test:

- Installed MPU6050 Arduino library.
- Wrote code to initialize the sensor.
- Tested communication using Serial Monitor.

Result:

- Arduino successfully detected the sensor.

Motion Detection Testing:

- Tested sensor by moving the module in different directions.

Result:

- Acceleration values changed when the sensor moved.
- The sensor correctly detects motion.

Conclusion:

- Core functionality works: motion data can be detected and read by the Arduino.

Prototype Demonstration Preparation:

- Prepared breadboard prototype for demonstration.
- Verified sensor communication and motion output.

Core Function Demonstrated:

- Motion detection
- Arduino reading sensor data
- Output displayed in Serial Monitor

Next:

- Convert motion data into cursor movement